

Problem Set 3

Version 1 (April 14, 2026)

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The problem set should be printed out and submitted in lecture. Please separately email me your code. You are encouraged to collaborate, but please write up your answers and work on computer code independently. Use any software you like, although you learn the most by avoiding canned statistics/econometrics packages and limiting the use of generative AI. If you find any errors or require clarification, please let me know right away.

Question 1

Assume the NKPC model from Question 6 of Problem Set 2. We maintain the assumption that inflation expectations are rational, so that $E_t(\pi_{t+1})$ is consistent with the true data generating process and with the agents' information set being given by current and past observations of π_t and x_t . Assume additionally: (i) $\gamma_b = 1 - \gamma_f$ and (ii) inflation follows the “autoregressive distributed lag” model

$$\pi_t = \mu_\pi + \rho_\pi \pi_{t-1} + \rho_x x_{t-1} + \eta_t,$$

where $(\nu_t, \eta_t)' \stackrel{i.i.d.}{\sim} (0_{2 \times 1}, \Sigma)$.

- i) Show that $(\pi_t, x_t)'$ follows a bivariate VAR(1) (you do not need to check that it is stationary). Derive explicitly how the four VAR coefficients (other than the intercepts) depend on the parameters $\theta \equiv (\gamma_f, \lambda, \rho_\pi, \rho_x)'$.
- ii) Using the same data as in Question 6 of Problem Set 2, run a least-squares VAR(1) regression on the data $(\pi_t, x_t)'$. Then use your answer to (i) to calculate a minimum distance estimate of θ based on these VAR estimates.
- iii) Report bootstrap confidence intervals for the parameters in θ (be explicit about your bootstrap procedure and the assumptions it relies on).

Question 2

Consider the univariate model

$$y_t = \mu + z_t, \quad (1)$$

where $\mu \in \mathbb{R}$ is unknown, and the unobserved mean-zero disturbance $\{z_t\}$ is strictly stationary and weakly dependent with unknown long-run variance. We are interested in doing inference on μ , so we use the sample average estimator $\hat{\mu} = \frac{1}{T} \sum_{t=1}^T y_t$. To form a confidence interval for μ based on $\hat{\mu}$, we need to estimate $\Omega \equiv \text{LRV}(z_t)$. We use a kernel HAC estimator

$$\hat{\Omega} \equiv \sum_{\ell=-(T-1)}^{T-1} k\left(\frac{\ell}{S_T}\right) \hat{\Gamma}(\ell), \quad \text{where } \hat{\Gamma}(\ell) \equiv \frac{1}{T} \sum_{t=\ell+1}^T (y_t - \hat{\mu})(y_{t-\ell} - \hat{\mu}).$$

We impose the Andrews (1991) assumptions on the kernel $k(\cdot)$, the bandwidth sequence S_T , and the time series properties of $\{z_t\}$. Then the bandwidth sequence that minimizes the asymptotic mean squared error (MSE) of $\hat{\Omega}$ in this one-dimensional problem is given by

$$S_T^* \equiv \left(\frac{q^* k_{q^*}^2}{\int_{-\infty}^{\infty} k(x)^2 dx} \alpha(q^*) T \right)^{1/(2q^*+1)},$$

where

$$\begin{aligned} \alpha(q^*) &\equiv \left(\frac{s_z^{(q^*)}(0)}{\Omega} \right)^2, \\ q^* &\equiv \sup\{q \geq 0 : k_q < \infty\}, \\ k_q &\equiv \lim_{x \rightarrow 0} \frac{1 - k(x)}{|x|^q}, \quad q \geq 0, \\ s_z^{(q)}(0) &\equiv \sum_{\ell=-\infty}^{\infty} |\ell|^q \Gamma_z(\ell), \quad q \geq 0. \end{aligned}$$

Here $\Gamma_z(\ell)$ denotes the autocovariance of $\{z_t\}$ at lag ℓ . Observe that the definition of $s_z^{(q)}(0)$ differs slightly from that in Andrews (1991), as I omit the constant $\frac{1}{2\pi}$.

Suppose we wish to implement the MSE-optimal bandwidth S_T^* in our one-dimensional model. To this end, we use the working assumption that $\{z_t\}$ follows a stationary AR(1) process

$$z_t = \rho z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2), \quad (2)$$

where $\rho \in (-1, 1)$, $\sigma^2 > 0$.

- i) Under the AR(1) assumption on $\{z_t\}$, derive an expression for the MSE-optimal HAC bandwidth S_T^* as a function of only T , ρ , and the kernel $k(\cdot)$.
- ii) Consider specifically the Bartlett and QS kernels, as defined in the “Frequentist inference” slide deck. For each of these two kernels, derive from first principles an expression for S_T^* as a function of only T and ρ . You may use without proof that when $k(\cdot)$ is the QS kernel (as it is defined in the slides), we have $\int_{-\infty}^{\infty} k(x)^2 dx = 1$, $q^* = 2$, and $k_2 = 1.421223$. [Hint: If you are stuck, you can look up the final expression in Andrews (1991) for inspiration. But you should still show the steps of the derivation in your assignment.]

Question 3

The data set `gk_data.csv` on the course website contains data from the paper “Monetary Policy Surprises, Credit Costs, and Economic Activity” by Gertler & Karadi (2015, *American Economic Journal: Macroeconomics* 7(1), 44–76).

The data is monthly and spans the time period 1990:1–2012:6. The time series are as follows:

<code>ipgr</code>	growth rate of industrial production, in annualized percentage points
<code>infl</code>	CPI inflation, in annualized percentage points
<code>gs1</code>	1-year government bond rate, in annualized percentage points
<code>ebp</code>	Excess Bond Premium (EBP), in annualized percentage points
<code>ff4_tc</code>	High-frequency changes in Federal Funds Futures prices

The variable `ebp` is a corporate bond spread that has been cleansed of default risk. The variable `ff4_tc` equals the monthly sum of high-frequency changes in three-month-ahead Federal Funds Futures prices measured in a short time window around Federal Open Market Committee (FOMC) monetary policy announcements (see Gertler & Karadi for details on the construction). `ff4_tc` equals 0 in months for which there is no FOMC meeting.

We seek to estimate the dynamic influence of monetary policy shocks on the EBP. We will use `ff4_tc` as a proxy/IV for the monetary shock. Collect the remaining four variables in the vector y_t . In the following, include 4 lags of the data and an intercept in all VARs and regressions.

Assume first that the monetary policy shock is invertible with respect to $\{y_t\}$, so that we can apply SVAR-IV.

- i) Estimate the impulse response function (IRF) of the EBP to a monetary policy shock, for horizons $\ell = 0, 1, \dots, 36$ months. Normalize the IRF such that it corresponds to a monetary shock that raises the 1-year government bond rate by 100 basis points on impact.
- ii) Estimate the Forecast Variance Decomposition (FVD) of the EBP with respect to the monetary policy shock, at forecast horizons $\ell = 1, 2, \dots, 36$ months.

Now we drop the invertibility assumption.

- iii) Estimate the IRF in (i) using LP-IV, implemented by ordering the IV first in a recursive VAR that also includes y_t .
- iv) Repeat (iii), but implement the LP-IV strategy using 2SLS regressions. Remember to control for lags of y_t and of the IV.
- v) Test the null hypothesis that the monetary policy shock is invertible, using the VAR specification in (iii).

Optional bonus question:

- vi) Provide 90% confidence intervals for your estimates in (i)–(iv). You may use any valid approach, but explain your assumptions. Plot the confidence intervals and point estimates together in an easily readable format.

Question 4

Go to the St. Louis Fed's FRED website and download data on industrial production (series code: INDPRO), personal consumption expenditures chained price index (PCEPI), and the constant maturity 1-year Treasury rate (GS1). Use the monthly series available on FRED. Only use data for the period January 1984 to December 2019.

Consider an SVAR in three variables:

1. Output growth, defined as the annualized log growth rate in INDPRO.
2. The PCE inflation rate, defined as the annualized log growth rate in PCEPI.
3. The real interest rate, defined as GS1 minus the inflation rate (GS1 is already measured in annualized percent).

Use 12 lags (and an intercept) when estimating the reduced-form VAR.

Let θ_ℓ denote the impulse response at horizon ℓ of output growth with respect to a *one-standard-deviation* contractionary monetary policy shock. To (partially) identify θ_ℓ , we impose the sign restrictions that a contractionary monetary policy shock raises the real interest rate and lowers the inflation rate at horizons $\ell = 0, 1, 2$. We do not restrict the other shocks in the model.

- i) Estimate the bounds of the identified set for θ_ℓ , separately for each $\ell = 0, 1, \dots, 36$. Do this by simulating (at least) one million rotation matrices Q from Haar measure,¹ retaining the ones that satisfy the sign restrictions given the estimated VAR, and then reporting the largest and smallest impulse responses of interest.
- ii) Now suppose we were subjective Bayesians who took Haar measure seriously as a prior for Q (truncated to the space where the sign restrictions hold). For simplicity, we will pretend that the estimated reduced-form VAR parameters are estimated without error, so the posterior uncertainty is only over Q . In this thought experiment, compute the posterior median and posterior equal-tailed 68% credible set for θ_ℓ , separately for each $\ell = 0, 1, \dots, 36$.
- iii) Plot the results of (i)–(ii) together in a single figure so that the various lines can be easily visually compared. Is the posterior median close to the center of the estimated identified set at every horizon?

¹If you simulate from Haar measure by using the QR decomposition as described in the slides, it is important to normalize the upper triangular R matrix to have positive diagonal.